Personalized Smartphone Usage Awareness System – Group 8

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1. Project Idea

Our project aims to develop a personalized Android application that anticipates intense or unwanted smartphone usage moments based on each user's own usage patterns. Rather than a warning system that applies the same static rules to everyone, our goal is a machine learning-based prediction model that adapts to each user's normal habits. The system will first be trained as a generalized model on a public dataset, and then personalized using data collected from the application's own users, thereby mitigating the "cold-start" problem. The feedback shown to users will be designed drawing on the behavior-change literature, particularly cognitive-behavioral approaches.

2. Literature Examples

Orzikulova et al. (2024), "Time2Stop: Adaptive and Explainable Human-AI Loop for Smartphone Overuse Intervention" (CHI 2024): proposed a system that predicts the most opportune moment for intervention using machine learning, provides explainable feedback, and adapts over time by feeding user responses back into the model; an 8-week field study with 71 participants achieved a 7–9% reduction in app-visit frequency. The authors explicitly addressed the "cold-start" problem in their discussion and highlighted fine-tuning a model pre-trained on a multi-user dataset with a small amount of new-user data as one solution — the generalized-plus-personalization architecture of our project directly implements this gap.

Pielot et al. (2017), "Beyond Interruptibility: Predicting Opportune Moments to Engage Mobile Phone Users" (UbiComp/IMWUT): predicted moments of user interruptibility using a machine learning model based on roughly 120 million raw smartphone usage events, and showed that incorporating a user's past behavior increased prediction accuracy fivefold compared to a basic baseline model. This finding is the direct literature foundation for our generalized-vs-personalized comparison, and the public dataset we use is also a product of this study.

3. Describe Your Data

In the initial phase we will use the public dataset published by Pielot et al. (2017), containing about 4 weeks of usage logs from 342 participants; the source study reported that a similarly scaled sample generated about 120 million events. The data is structured as one CSV file per participant, containing screen on/off and unlock events, foreground app, notification events, location, and timestamp information. Since the dataset has no ready-made "unwanted usage" label, we will engineer the target variable ourselves using a proxy definition built from exceptionally long continuous sessions, late-night usage, and rapid consecutive unlock events. Preprocessing will divide the event logs into fixed 30-minute windows, extract features such as screen time, unlock count, app-switch count, and previous session duration for each window, and assign the proxy label to these windows; collecting fine-grained scrolling dynamics is left to later stages as an extension the team will evaluate. In subsequent stages, data collected from our application's own users will serve as the second data source, used for personalization.

4. Approach (Machine Learning or Deep Learning)

A machine learning approach will be preferred, specifically tree-based algorithms such as XGBoost and Random Forest. Because features are extracted per window and are numerical and relatively low-dimensional, such methods are more efficient than deep learning at this scale and structure, and they also yield interpretable results via feature importance — aligning with the explainability goal emphasized by Time2Stop. Three models will be compared: (i) a generalized model trained solely on the public dataset, (ii) a personalized model adapted from the generalized model with personal data, and (iii) a model trained from scratch solely on personal data. This comparison will directly test our cold-start hypothesis by measuring after how many days of personal data personalization begins to make a noticeable difference.